INCH-POUND

MIL-E-62462B(AT)
AMENDMENT 1
30 November 1992

MILITARY SPECIFICATION

ENGINE, DIESEL, LIQUID-COOLED, 8 CYLINDER;
4 CYCLE, V-TYPE, 295 H.P. (220 KW)

This amendment forms a part of MIL-E-62462B(AT), dated 26 August 1985, and is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

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2.1.1, SPECIFICATIONS, FEDERAL: delete "O-I-490 -Inhibitor, Corrosion, Liquid Cooling System."

2.1.1, SPECIFICATIONS, MILITARY: after MIL-A-46167, add "MIL-A-53009 -Additive, Antifreeze Extender, Liquid Cooling System.", add "MIL-T-83133 -Turbine Fuel, Aviation, Kerosene Type, Grade JP-8."

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3.3.1.1, line 1: delete "a satisfactory", and substitute "an".

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3.3.1.4, subparagraph c.: delete "127" and substitute "150"; delete "481" and substitute "568".

3.3.1.6 Delete in its entirety and substitute:

"3.3.1.6 Coolant. The engine shall perform as specified herein using one of the following coolants as required for the operational or storage/transit conditions shown in table I-1:

a. Water and additives conforming to MIL-A-53009.
b. Water and antifreeze compound conforming to O-A-548.
c. Arctic-type antifreeze conforming to MIL-A-11755."

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3.3.1.7, subparagraph b, delete in its entirety and substitute:

"b. Coolant:
-Maximum continuous, 210°F (99°C) (measured at engine outlet).
-Maximum intermittent, 230°F (110°C) not to exceed 15 minutes in duration (measured at inlet to transmission oil cooler)."

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3.3.1.10.1 Delete in its entirety and substitute:

"3.3.1.10.1 Fuel. The engine shall perform as specified herein using DF-2 grade fuel in accordance with VV-F-800. The engine shall be capable of operating over the entire temperature range specified herein using JP-8 grade fuel in accordance with MIL-T-83133. DF-A, DF-1, or JP-8 shall be required for low temperature ambient conditions as specified herein (see 4.6.1.6)."

3.3.1.13 Add:

"3.3.1.13 50 hour performance. The engine shall be capable of operating at the specified speeds and loads in 4.6.1.11 while meeting the following:

a. Performance characteristics of figure 1.
b. Oil sump temperature (see 3.3.1.7a).
c. Coolant temperature (see 3.3.1.7b)."

3.3.4, line 2: delete "immediately .... herein and" and substitute "after the climatic conditions specified in table I-1 when tested".

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Table I-1, Add:

TABLE I-1. Temperature, solar radiation and relative humidity.

Climatic Design Type	Operational Conditions: Daily Cycle (QSTAG 360 equiv.)*	Operational Conditions: Ambient Air Temp °F (°C)*	Operational Conditions: Solar Radiation BPH (W/m)*	Operational Conditions: Ambient Relative Humidity %	Storage and Transit Conditions: Induced Air Temp °F (°C)*	Storage and Transit Conditions: Induced Relative Humidity %
Hot	Hot-Dry (A1)	90to120 (32to49)	0 to 355 (0to1120)	3 to 8	91to160 (33to71)	1 to 7
	Hot-Humid (B3)	88to105 (31to41)	0 to 243 (0to1080)	59 to 88	91to160 (33to71)	14 to 80
	Constant High Humidity (B1)	Nearly Constant 75 (24)	Negligible	95to100	Nearly Constant 80 (27)	95to100
Basic	Variable High Humidity (B2)	78to95 (26to35)	0 to 37 (0to970)	74to100	86to145 (30to63)	19 to 75
	Basic Hot (A2)	86to110 (30to43)	0 to 355 (0to1120)	14 to 44	86to145 (30to63)	5 to 44
	Basic Cold (C1)	-5to-25 (-21to-32)	Negligible	Tending Toward Saturation	-13to-28 (-25to-33)	Tending Toward Saturation

*Designations in parenthesis refer to corresponding climatic condition in Quadripartite Standardization Agreement 360 Climatic Environmental Conditions Affecting the Design of Military Materiel.

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4.4.2, Delete the first sentence and substitute:

"Unless otherwise specified (see 6.2), the Government shall select two units for initial production inspection from the first ten engines produced under the produciton contract."

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Table II, Item 207: delete "none in threaded holes.", and substitute "paint in threaded holes."

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4.5.3.2, Delete in its entirety and substitute:

"4.5.3.2 Control tests. Control tests shall be conducted on one randomly selected engine assembly from each lot of 60 units consecutively produced, except at least one test shall be performed in a six month period. The engine assemblies shall be subjected to the test specified in table III."

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4.5.4 NOTE, line 2: delete "or endurance", and substitute "endurance, or 50 hour perfomance".

4.5.4 NOTE, line 7, after "permitted": add "Prior to and subsequent to the above tests an oil sample shall be taken to verify contamination does not exceed the manufacturer's limits."

4.5.4 NOTE, line 11: delete "break-in-run and".

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Table III, Control test column: delete all X's in column.

Table III, Acceptance test column: add "X" across from 3.3.1.8, 3.3.1.10, and 3.3.2.

Table III, after "Endurance": add "50 hour performance 3.3.1.13 4.6.1.11", and add an "X" in Control column.

4.6.1.11 Add:

"4.6.1.11 50 Hour performance. To determine conformance to 3.3.1.13, verify the engine performance capabilities conform to the maximum curves specified in figure 1 under the following conditions:

Duration (hrs)	Engine speed (rpm)	Load (%)
5	1500	50
5	1850	85
5	2600	0
5	2250	85
30	2600	100"

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4.6.4, Delete in its entirety and substitute:

"4.6.4 Enviromental conditions. To determine conformance to 3.3.4, the engine shall be tested in accordance with MIL-STD-810.

a. Method 501.2, High Temperature Operation, Procedure II, Cyclic. Incorporate the following cyclic conditions into Procedure II:

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-Raise the internal chamber temperature to 120°F (49°C).
-Maintain internal chamber temperature for 6 hours at 120°F (49°C).
-Raise the internal chamber temperature to 160°F (71°C) within a time period of 1 hour and then maintain at that temperature for 4 additional hours.
-Lower the internal chamber temperature to 120°F (49°C) within a time period of 1 hour.
-Repeat steps 3, 4, and 5 two additional times (making a total of three 12-hour cycles).
-Adjust the internal chamber temperature to the highest operating temperature under which the test item is designed to operate and maintain until temperature stabilization of the test item is reached.

b. Method 502.2 Low Temperature Operation, Procedure II. Incorporate the following into procedure II:

-Prepare for test.
-Lower the internal chamber temperature to the storage temperature -28°F (-33°C) as specified in the equipment specification and maintain for a period of 24 hours after stabilization or for the period specified in the equipment specification.
-Inspect.
-Adjust the internal chamber temperature to -25°F (-32°C) and maintain until temperature stabilization of the test item is reached.

6.2, subparagraph e, delete in its entirety and substitute:

"e. Control tests, if required (see 4.5.3.2).
f. Selection of applicable levels of preservation, packaging, packing and marking (see 5.1)."

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Figure 1, add to the bottom of page:

"NOTE: Maximum performance curves are nominal within a ± 5% tolerance band."

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